

Michelle Zuo

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EDUCATION

Duke University

Aug. 2026 – May 2030

- B.S. Candidate in Mathematics and Computer Science

Thomas Jefferson High School for Science and Technology

Graduated Jun. 2026

- Weighted GPA: 4.589

RESEARCH & TECHNICAL EXPERIENCE

Research Intern, Aspiring Scientists Summer Internship Program

Jun. 2025 – Jun. 2026

- Analyzed relationships between economic loss from NOAA's 10 costliest disaster hazards and social vulnerability to inform disaster-aid allocation.
- Processed terabyte-scale geospatial data with Python and GeoDa; built statistical models and spatial visualizations. Co-authored a paper under journal review.

Research Student, Institute for Mathematics and Democracy

Jun. 2025 – Apr. 2026

- Applied Möbius inversion and machine-learning models to Wisconsin election data on a 3-person team; presented findings and continued collaboration with faculty mentor.
- Authored *Analyzing Swing States in Presidential Elections: The Case of Wisconsin*, an arXiv preprint.

Quantum Researcher, Virginia Tech Partnership

Sep. 2025 – Jun. 2026

- Researched ADAPT-VQE, an adaptive variational quantum algorithm for molecular simulation, using Python and quantum-entanglement methods under Virginia Tech faculty mentorship.

LEADERSHIP

Founding Officer & Tutor, TJ Math Center

Aug. 2024 – Jun. 2026

- Recruited and trained 90+ peer tutors; managed 72 weekly tutoring slots supporting 1,000+ sessions.
- Tutored 50+ students in precalculus, AP Calculus AB/BC, and statistics.

Programs Director & Teacher, Girls Who Math

May 2022 – Jun. 2026

- Led 7 instructors and 7 teaching assistants delivering math, science, and Python/Java instruction to 400–500 students annually.
- Developed curricula across seven subjects and taught 100+ middle-school students nationwide.

Officer, Varsity Math Team

Aug. 2022 – Jun. 2026

- Led weekly training for 80+ students and organized TJIMO for 150+ participants; placed 2nd in Duke Math Meet Guts Round and 6th in CMIMC TCS Round.

Publicist & Member, TJHSST Math Modeling Club

Aug. 2023 – Jun. 2026

- Earned M3 Challenge Successful in Triage distinction (top 18% internationally); increased club participation 50% through outreach.

Senator, Class Council

Jun. 2024 – Jun. 2026

- Represented the Class of 2026 as a liaison to faculty; organized school-wide events and assemblies; raised \$27,000 for prom through fundraisers and student initiative; led 4 homecoming committees with 40+ volunteers annually.

AWARDS

Mathematics & Computing: AIME Qualifier (x4); AIME Honor Roll, top 5% (2x); NCWIT Aspirations in Computing State Winner (2026) and Honorable Mention (2025); M3 Challenge Successful in Triage, top 18% internationally (2025, 2026).

Research & Academic: 1st Place, Fairfax County Regional Science & Engineering Fair — Energy: Sustainable Materials and Design (2023); AP Scholar with Distinction (2025, 2026); National Merit Commended Scholar (2025).

TECHNICAL SKILLS

Languages: Python, Java, R, MATLAB, LaTeX

Libraries: pandas, NumPy, Matplotlib, Qiskit, PyTorch